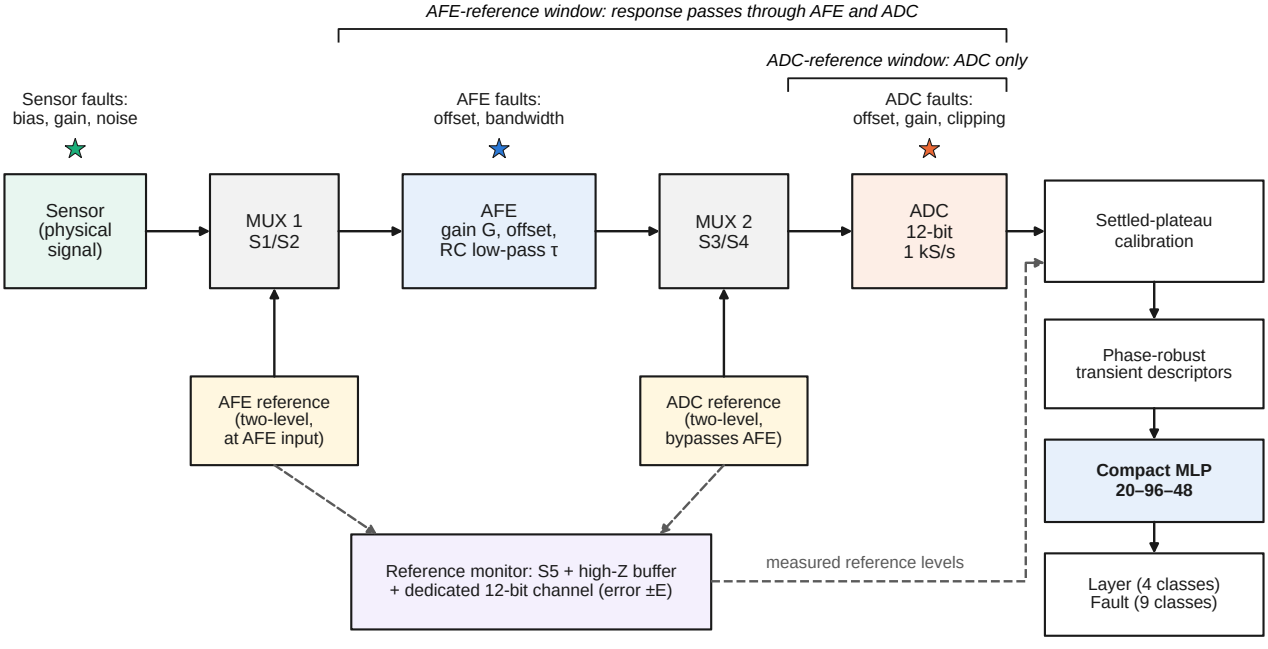
AFE-reference window: response passes through AFE and ADC
ADC-reference window: ADC only
Sensor faults:
bias, gain, noise
AFE faults:
offset, bandwidth
ADC faults:
offset, gain, clipping
Sensor
(physical
signal)
MUX 1
S1/S2
AFE
gain G, offset,
RC low-pass τ
MUX 2
S3/S4
ADC
12-bit
1 kS/s
Settled-plateau
calibration
Phase-robust
transient descriptors
Compact MLP
20–96–48
Layer (4 classes)
Fault (9 classes)
AFE reference
(two-level,
at AFE input)
ADC reference
(two-level,
bypasses AFE)
Reference monitor: S5 + high-Z buffer
+ dedicated 12-bit channel (error ±E)
measured reference levels